Name
Date
Course

Applications of Algebra (Part III)

Special Bases

Base "e"

"e" is an irrational number (i.e. non-repeating, non-terminating). Its value is below.

$$e = 2.718281828845...$$

"e" is special because it is used in many applications, including finance and Science.

Because base "e" powers are used so frequently, base "e" logarithms are abbreviated as follows:

$$\log_e X = \ln X$$

"ln" is short for "natural logarithm." Base "e" logarithms are given this name because they are used in real-world applications (e.g. biology, chemistry, physics, finance, etc.).

More Examples

$$\log_e 2 = \ln 2$$

$$\log_e\left(\frac{8}{7}\right) = \ln\left(\frac{8}{7}\right)$$

Base Ten

Base ten powers are special because it is easy to perform calculations with them. Consequently, they are used in many applications, including science (e.g. measuring the loudness of sound).

Because base ten powers are used so frequently, base ten logarithms are abbreviated as follows:

$$\log_{10} X = \log X$$

More Examples

$$\log_{10} 3 = \log 3$$

$$\log_{10} 8.2 = \log 8.2$$

Practice

Write the abbreviated forms of the logarithms below.

① $\log_{10} 562 = \boxed{\log 562}$ ② $\log_e 0.9 = \boxed{\ln 0.9}$ ③ $\log_{10} \pi = \boxed{\log \pi}$

④ $\log_e\left(\frac{4}{y}\right) = \boxed{\ln\left(\frac{4}{y}\right)}$ ⑤ $\log_e 8{,}000 = \boxed{\ln 8{,}000}$ ⑥ $\log_{10}\left(\frac{2}{3}\right) = \boxed{\log\left(\frac{2}{3}\right)}$

⑦ $\log_e x^2 = \boxed{\ln x^2}$ ⑧ $\log_{10}(3y) = \boxed{\log(3y)}$

Solving Exponential Equations with Special Bases

Solve the following equations. Write the Roman numeral of each exponent rule that you use in the margin. Round to the nearest thousandth if necessary.

⑨ $7 = e^x \cdot e^{3x}$

I $7 = e^{x+3x}$

$7 = e^{4x}$

$4x = \log_e 7$

$4x = \ln 7$

$x = \frac{\ln 7}{4}$

$x \approx \frac{1.9459}{4} \approx \boxed{0.486}$

⑩ $\frac{10^{7x}}{10^{5x}} = 8$

II $10^{7x-5x} = 8$

$10^{2x} = 8$

$2x = \log_{10} 8$

$2x = \log 8$

$x = \frac{\log 8}{2} \approx \frac{0.903}{2} \approx \boxed{0.452}$

⑪ $\dfrac{e^{12x}}{e^{7x}}=3$

II $e^{12x-7x}=3$

$e^{5x}=3$

$5x=\log_e 3$

$5x=\ln 3$

$x=\dfrac{\ln 3}{5}$

$x\approx\dfrac{1.09861}{5}$

$\boxed{x\approx 0.220}$

⑫ $10^{x}\cdot 10^{2x}=18$

I $10^{x+2x}=18$

$10^{3x}=18$

$3x=\log_{10}18$

$3x=\log 18$

$x=\dfrac{\log 18}{3}$

$x\approx\dfrac{1.25527}{3}$

$\boxed{x\approx 0.418}$

⑬ $-8=1-2e^{x}$

$-9=-2e^{x}$

$\dfrac{-9}{-2}=\dfrac{-2e^{x}}{-2}$

$\dfrac{9}{2}=e^{x}$

$x=\log_e\left(\dfrac{9}{2}\right)$

$x=\ln\left(\dfrac{9}{2}\right)\approx\boxed{1.504}$

⑭ $6(10^{x})+4=16$

$6(10^{x})=12$

$\dfrac{6(10^{x})}{6}=\dfrac{12}{6}$

$10^{x}=2$

$x=\log_{10}2$

$x=\log 2$

$\boxed{x\approx 0.301}$

⑮ $3e^{x}-1=11$

$3e^{x}=12$

$\dfrac{3e^{x}}{3}=\dfrac{12}{3}$

$e^{x}=4$

$x=\log_e 4=\ln 4\approx\boxed{1.386}$

⑯ $14=7+5(10^{x})$

$7=5(10^{x})$

$\dfrac{7}{5}=\dfrac{5(10^{x})}{5}$

$\dfrac{7}{5}=10^{x}$

$x=\log_{10}\left(\dfrac{7}{5}\right)=\log\left(\dfrac{7}{5}\right)\approx\boxed{0.146}$

Solving Logarithmic Equations with Special Bases

Solve the following equations. Write the Roman numeral of each exponent rule or logarithm property that you use in the margin. Round to the nearest thousandth if necessary.

⑰ $2\log x-5=5$

$2\log x=10$

$\log x=5$

$\log_{10}x=5$

$10^{5}=x$

$\boxed{100{,}000=x}$

⑱ $13=-\dfrac{\ln x}{3}+15$

$-2=-\dfrac{\ln x}{3}$

$(-3)(-2)=-\dfrac{\ln x}{3}(-3)$

$6=\ln x$

$6=\log_e x$

$x=e^{6}\approx\boxed{403.429}$

(19) $-17 = 4\ln x - 21$

$4 = 4\ln x$

$1 = \ln x$

$1 = \log_e x$

$e^1 = x$

EVI $x = e$

$\boxed{x \approx 2.718}$

(20) $\frac{\log x}{5} - 1 = -1$

$\frac{\log x}{5} = 0$

$\log x = 0$

$\log_{10} x = 0$

$10^0 = x$

EVIII $\boxed{1 = x}$

Logarithm Property V

$\log_k(AB) = \log_k A + \log_k B$

Logarithm Property VI

$\log_k\left(\frac{A}{B}\right) = \log_k A - \log_k B$

(21) $\log 2 + \log x = 1$

LI $\log_{10} 2 + \log_{10} x = 1$

$\log_{10}(2x) = 1$

$10^1 = 2x$

EVI $10 = 2x$

$\boxed{x = 5}$

(22) $3 = \ln x - \ln 7$

$3 = \log_e x - \log_e 7$

LVI $3 = \log_e\left(\frac{x}{7}\right)$

$e^3 = \frac{x}{7}$

$x = 7e^3$

$x \approx 7(20.085) \approx \boxed{140.599}$

(23) $3 = \log x - \log 8$

LVII $3 = \log_{10} x - \log_{10} 8$

$3 = \log_{10}\left(\frac{x}{8}\right)$

$10^3 = \frac{x}{8}$

$1{,}000 = \frac{x}{8}$

$\boxed{x = 8{,}000}$

(24) $\ln 3 + \ln x = \frac{1}{2}$

$\log_e 3 + \log_e x = 1/2$

LV $\log_e(3x) = 1/2$

$e^{1/2} = 3x$

EIX $\sqrt{e} = 3x$

$x = \frac{\sqrt{e}}{3} \approx \frac{1.6487}{3} \approx \boxed{0.55}$

Logarithm Property VII

$$\log_K A^B = B\log_K A$$

(25) $4\log X + \log 9 = 2$

$4\log_{10}X + \log_{10}9 = 2$

LVII $\log_{10}X^4 + \log_{10}9 = 2$

LVI $\log_{10}(9X^4) = 2$

$10^2 = 9X^4$

$100 = 9X^4$

$\frac{100}{9} = X^4$

$\pm\sqrt[4]{\frac{100}{9}} = X$

$X = \sqrt[4]{\frac{100}{9}}$ or $X \neq -\sqrt[4]{\frac{100}{9}}$

$\boxed{X \approx 1.826}$

(26) $\ln 10 - \frac{1}{3}\ln X = 2$

$\log_e 10 - \frac{1}{3}\log_e X = 2$

LVII $\log_e 10 - \log_e X^{\frac{1}{3}} = 2$

LVI $\log_e\left(\frac{10}{X^{1/3}}\right) = 2$

$e^2 = \frac{10}{X^{1/3}}$

$X^{1/3} \cdot e^2 = \frac{10}{X^{1/3}} \cdot X^{1/3}$

$e^2 X^{1/3} = 10$

$\frac{e^2 X^{1/3}}{e^2} = \frac{10}{e^2}$

$X^{1/3} = \frac{10}{e^2}$

EIX $\sqrt[3]{X} = \frac{10}{e^2}$

$X = \left(\frac{10}{e^2}\right)^3 \approx \boxed{2.479}$

(27) $3\log X + \log 2 = 2$

$3\log_{10}X + \log_{10}2 = 2$

LVII $\log_{10}X^3 + \log_{10}2 = 2$

LVI $\log_{10}(2X^3) = 2$

$10^2 = 2X^3$

$100 = 2X^3$

$\frac{100}{2} = \frac{2X^3}{2}$

$50 = X^3$

$\sqrt[3]{50} = X$

$\boxed{X \approx 3.684}$

(28) $2\ln X - \ln 6 = 0$

$2\log_e X - \log_e 6 = 0$

LVII $\log_e X^2 - \log_e 6 = 0$

LVII $\log_e\left(\frac{X^2}{6}\right) = 0$

$e^0 = \frac{X^2}{6}$

EVII $1 = \frac{X^2}{6}$

$6 = X^2$

$\pm\sqrt{6} = X$

$X = \sqrt{6}$ or $X \neq -\sqrt{6}$

$\boxed{X \approx 2.449}$

Applications of Base Ten Logarithms

pH: A measure of the acidity of a liquid and defined as $-\log[H^+]$, where $[H^+]$ is the concentration of hydrogen ions in the liquid. $[H^+]$ is measured in moles (i.e. # of molecules) per liter.

Each number below represents the hydrogen ion concentration of a liquid. Find the pH of each liquid.

(29) $[H^+] = 0.005$ moles/liter

$$pH = -\log[H^+] = -\log_{10}(0.005) \approx -(-2.3)$$
$$\approx \boxed{2.3}$$

(30) $[H^+] = 0.00002$ mole/liter

$$pH = -\log[H^+] = -\log_{10}(0.00002) \approx -(-4.699)$$
$$\approx \boxed{4.699}$$

(31) $[H^+] = 0.000000049$ moles/liter

$$pH = -\log[H^+] = -\log_{10}(0.000000049) \approx -(-7.3)$$
$$\approx \boxed{7.3}$$

(32) $[H^+] = 0.0031$ moles/liter

$$pH = -\log[H^+] = -\log_{10}(0.0031) \approx -(-2.5)$$
$$\approx \boxed{2.5}$$

Each number below represents the pH of a liquid. Find the hydrogen ion concentration of each liquid.

(33) $pH = 8$

$$pH = -\log[H^+]$$
$$8 = -\log[H^+]$$
$$8 = -\log_{10}[H^+]$$
$$(-1)8 = (-\log_{10}[H^+])(-1)$$
$$-8 = \log_{10}[H^+]$$
$$10^{-8} = [H^+]$$
$$[H^+] = \boxed{0.00000001 \text{ moles/liter}}$$

(34) $pH = 5.71$

$$pH = -\log[H^+]$$
$$5.71 = -\log[H^+]$$
$$-5.71 = \log_{10}[H^+]$$
$$10^{-5.71} = [H^+]$$
$$[H^+] \approx \boxed{0.00000195 \text{ moles/liter}}$$

(35) $pH = 2$

$$pH = -\log[H^+]$$
$$2 = -\log[H^+]$$
$$-2 = \log[H^+]$$
$$-2 = \log_{10}[H^+]$$
$$10^{-2} = [H^+]$$
$$[H^+] = \boxed{0.01 \text{ moles/liter}}$$

(36) $pH = 10.3$

$$pH = -\log[H^+]$$

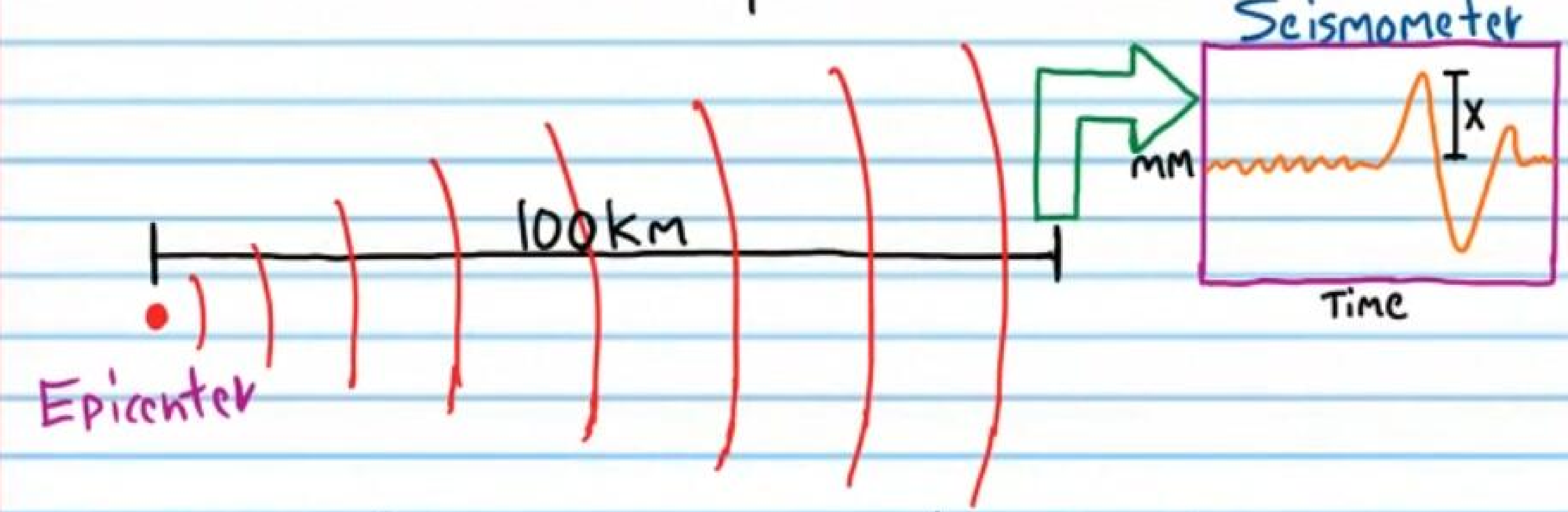
$$10.3 = -\log_{10}[H^+]$$

$$-10.3 = \log_{10}[H^+]$$

$$10^{-10.3} = [H^+]$$

$$\boxed{[H^+] = 0.0000000000501 \text{ moles/liter}}$$

The Magnitude M of an Earthquake: The magnitude of an earthquake is defined as $\log(1000x)$, where x is the length of a disturbance (in millimeters) on a seismometer 100 kilometers from the epicenter.



The following numbers are the lengths of disturbances on a seismometer 100 kilometers from the epicenters of earthquakes. Find the magnitude of each earthquake.

(37) 0.1 millimeters

$$M = \log(1000x) = \log_{10}[1000(0.1)] = \log_{10} 100 = \boxed{2}$$

(38) 8 millimeters

$$M = \log(1000x) = \log_{10}[1000(8)] = \log_{10} 8000 \approx \boxed{3.9}$$

(39) 40,000 millimeters

$$M = \log(1000x) = \log_{10}[1000(40{,}000)] = \log_{10}[40{,}000{,}000]$$
$$\approx \boxed{7.6}$$

(40) 178 millimeters

$$M = \log(1000x) = \log_{10}[1000(178)] = \log_{10}(178{,}000)$$
$$= \boxed{5.25}$$

(41) 10 millimeters

$$M = \log(1000x) = \log[1000(10)] = \log(10{,}000)$$
$$= \boxed{4}$$

(42) 5,000,000 millimeters

$$M = \log(1000x) = \log[1000(5{,}000{,}000)]$$
$$= \log[5{,}000{,}000{,}000]$$
$$\approx \boxed{9.7}$$

Find the lengths of disturbances on a seismometer 100 kilometers from the epicenters of various earthquakes if the numbers below represent magnitudes of each earthquake.

43 $M = 6.3$

$$M = \log(1000x)$$

$$6.3 = \log_{10}(1000x)$$

$$10^{6.3} = 1000x$$

$$1{,}995{,}262 \approx 1000x$$

$$\boxed{x \approx 1{,}995.262 \text{ millimeters}}$$

44 $M = 3.0$

$$M = \log(1000x)$$

$$3 = \log_{10}(1000x)$$

$$10^{3} = 1000x$$

$$1000 = 1000x$$

$$\boxed{x = 1 \text{ millimeter}}$$

45 $M = 4.8$

$$M = \log(1000x)$$

$$4.8 = \log_{10}(1000x)$$

$$10^{4.8} = 1000x$$

$$63{,}095.7 \approx 1000x$$

$$\boxed{63.1 \text{ millimeters} \approx x}$$

46 $M = 10$

$$M = \log(1000x)$$

$$10 = \log_{10}(1000x)$$

$$10^{10} = 1000x$$

$$10{,}000{,}000{,}000 = 1000x$$

$$\boxed{x = 10{,}000{,}000 \text{ millimeters}}$$